MATH 302: Homework 1

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Question 1

Question. Show that every normed vector space $(V, \|\cdot\|)$ is automatically a metric space (V, d) with the metric d given by $d(x, y) = \|x - y\|$.

Definition (1.1 Metric Space Properties) . A set is a metric space if it is equipped with a function $d : V \times V \to \mathbb{R}$ such that for all $x, y, z \in V$:

Property	**Description**
I. Identity of Indiscernibles	$d(x, y) = 0 \Leftrightarrow x = y$
II. Symmetry	$d(x, y) = d(y, x)$
III. Triangle Inequality	$d(x, y) \leq d(x, z) + d(z, y)$
IV. Non-Negatitivity	$d(x, y) \geq 0$

Let there be a normed vector space $(V, \|\cdot\|)$. Define $d(x, y) = \|x - y\|$. We will show that d satisfies the four properties of a metric space.

Identity of Indiscernibles

Observe that the "Identity of Indiscernibles" property states that $d(x, y) = 0$ if and only if $x = y$. In other words given $x, y \in V$ and the distance between them is zero, then they must be the same point.

By the definition of d, we have:

$$d(x, y) = \|x - y\|$$

Therefore if $x < y$ or $y < x$ then $0 < \|x - y\|$. By trichotomy, we have that $x = y$ if and only if $\|x - y\| = 0$. Thus, the "Identity of Indiscernibles" property holds. ■

Symmetry

Observe the "Symmetry" property holds if the distance from x to y is the same as the distance from y to x. In other words, given $x, y \in V$, we have that $d(x, y) = d(y, x)$.

We can test this assertion by substituting the definition of d into the equation:

$$d(x, y) = \|x - y\| = \|-(y - x)\| = \|y - x\| = d(y, x)$$

Thus, the "Symmetry" property holds. ■

Triangle Inequality

Observe the "Triangle Inequality" property states the distance between two points x and y is less than or equal to the sum of the distances from x to a third point z and from z to y. In other words, given $x, y, z \in V$, we have that $d(x, y) \leq d(x, z) + d(z, y)$.

We can test this assertion by substituting the definition of d into the equation:

$$d(x, z) = \|x - z\| \wedge d(z, y) = \|z - y\| \wedge d(x, y) = \|x - y\|$$
$$d(x, z) + d(z, y) = \|x - z\| + \|z - y\|$$

Observe the following:

$$\|x - y\| = \|(x - z) + (z - y)\| \leq \|x - z\| + \|z - y\|$$

Thus, the "Triangle Inequality" property holds. ■

Non-Negativity

Observe the "Non-Negativity" property states that the distance between two points x and y is always greater than or equal to zero. In other words, there can never be a negative distance between two points.

As the distance function d is defined as the norm of the difference between two points, we have that $d(x, y) = \|x - y\|$. By the definition of a norm, we know that $\|v\| \geq 0$ for all $v \in V$. Therefore, $d(x, y) \geq 0$ for all $x, y \in V$. ■

Conclusion

As we have shown that d satisfies all four properties of a metric space, we conclude that every normed vector space $(V, \|\cdot\|)$ is automatically a metric space (V, d) with the metric d given by $d(x, y) = \|x - y\|$.

Question 2

Question. On $\mathbb{R}^n$ define, for any $x = (x_1, ..., x_n)$,

$$\|x\|_1 = \sum_{j=1}^{n} |x_j| \quad \text{and} \quad \|x\|_\infty = \max\{|x_1|, ..., |x_n|\}.$$

Show that both $\|x\|_1$ and $\|x\|_\infty$ are norms on $\mathbb{R}^n$.